



Numerical Predictions of Bottom Layer Stability in Material Extrusion Additive Manufacturing

MD TUSHER MOLLAH ^{1,2,3} RAPHAËL COMMINAL ¹
MARCIN P. SERDECZNY ¹ DAVID B. PEDERSEN ¹
and JON SPANGENBERG ^{1,4}

1.—Department of Mechanical Engineering, Technical University of Denmark, 2800 Kgs. Lyngby, Denmark. 2.—e-mail: mtumo@mek.dtu.dk. 3.—e-mail: tusher.bsmrstu@gmail.com. 4.—e-mail: josp@mek.dtu.dk

Robocasting and 3D concrete printing are technologies that belong under the umbrella term material extrusion additive manufacturing. These two free-form fabrication methods are used to produce 3D structures/components in materials such as ceramic pastes, thermosets, and concrete. Common for the materials is their viscoplastic behavior during deposition and structural buildup (i.e., increase in yield stress) after deposition. The material's complex nature makes it a non-trivial task to ensure that printed layers do not deform when depositing additional layers on top. In this article, we numerically investigate the influence of the yield stress buildup of viscoplastic materials on the stability of the bottom layer during multilayer printing. Specifically, we have developed a computational *flssssuid* dynamics model that applies a scalar approach to alter the yield stress. The novel model provides fundamental knowledge on how to design the material's rheology, so the bottom layer can withstand both the hydrostatic and extrusion pressure.

INTRODUCTION

Recently, Material Extrusion Additive Manufacturing (MEX-AM) of viscoplastic materials has received more attention when trying to produce micro-components and large-scale structures.¹ Viscoplastic materials such as ceramic pastes, thermosets, and concrete are typically used in free-form fabrication technologies such as 3D concrete printing and Robocasting, which belong under the umbrella term MEX-AM.^{1–3} In these technologies, a layer-by-layer approach is used to fabricate the structure. The material is extruded through a nozzle and deposited onto a solid support for the first layer, while the following layers are deposited onto the already printed layer.⁴ The already printed layer can be deformable (i.e., wet or partially solidified) or solidified. When printing on a solidified material, it is easy to obtain a stable print (i.e.,

layers with equal width and height); however, it can have detrimental consequences on the interlayer strength of the fabricated parts.⁵ Consequently, it is often preferable to print on a deformable layer although it is non-trivial to obtain a stable print.

The geometrical accuracy and stability of the prints in MEX-AM require significant consideration.^{6–8} Numerical models have with success been applied to improve various manufacturing processes.⁹ Specifically, Computational Fluid Dynamics (CFD) models have shown to be a very beneficial tool to understand MEX-AM, e.g., improve the quality of printed corners,^{10,11} reducing porosities,¹² and increase the inter- and intra-layer bonding of deposited layers.¹³ Other CFD models^{14–16} have shown that the extruded material also applies a pressure, called the extrusion pressure, to the closest surface (solid/fluid) during the deposition. Comminal et al.¹⁷ Roussel et al.¹⁸ and Spangenberg et al.¹⁹ all discussed the consequences of extrusion pressure on the stability of a deposited layer in MEX-AM. Recently, Mollah et al.²⁰ investigated how to reduce the extrusion pressure and its effect

(Received September 13, 2021; accepted November 8, 2021)

on the stability of the structure. However, the stability of the structure during multilayer prints is still to be fully understood.

This work aims to numerically investigate the stability of layers during multilayer printing of viscoplastic materials in MEX-AM. Specifically, a two-layered structure is investigated where the second layer is printed onto a partially solidified bottom layer. Numerically this is obtained by assigning a different yield stress for the bottom layer. The CFD model is used to predict the cross-sectional shapes of deposited layers and analyze the deformation of the bottom layer as well as the extrusion pressure.

COMPUTATIONAL MODEL

The CFD model simulates the extrusion-deposition flow of the viscoplastic materials that is assumed to be incompressible with a constant density ρ . The deposition flow properties are laminar, time dependent and isothermal. The fluid is introduced with a fully developed profile at an extrusion speed of 40 mm/s. The material is deposited through a cylindrical nozzle with a 25 mm diameter onto a solid substrate located at a distance 12.5 mm from the nozzle exit. The nozzle moves with a printing speed of 50 mm/s. The model geometry along with the computational domain is presented in Fig. 1. Only half of the domain is solved by imposing a symmetry plane. An inlet boundary is assigned to the top plane of the domain designed with an artificial solid component that coincides with the nozzle orifice to prevent flow from the rest of the plane. The bottom plane is a stationary wall having the substrate. The rest of the planes have no interaction with the fluid and have assigned a continuative boundary condition that considers a null derivative of the velocity profile across the plane. A no-slip boundary condition is assigned to

the solid surfaces that are in contact with the fluid. The computational domain is meshed with the Cartesian grid. The cell size is defined to have 25 cells along the diameter of the nozzle.

The viscosity of the material is modeled by the bi-viscous regularization²¹ of the Bingham constitutive equation.²² The apparent viscosity of the material μ_{ap} is:

$$\mu_{ap}(\dot{\gamma}) = \begin{cases} \mu_{M.ap} = \eta_P + \frac{\tau_{printing}}{\dot{\gamma}_c}, & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \eta_P + \frac{\tau_{printing}}{\dot{\gamma}}, & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (1)$$

where $\mu_{M.ap}$ is the maximum apparent viscosity for the shear rates $\dot{\gamma}$ below a critical value $\dot{\gamma}_c$, $\tau_{printing}$ is the yield stress, and η_P is the plastic viscosity. A time-dependent yielding behavior of the material is considered. This means that the yield stress of previously deposited material is assumed to increase before the deposition of the following layer begins. Numerically, the yield stress difference between the layers is introduced by the scalar approach of FLOW-3D[®], version 12.0.²³ This is done by assigning a constant scalar concentration to the already deposited material. The scalar carries a viscosity that is modeled, with a different yield stress $\tau_{printed}$, according to Eq. 1. The apparent viscosity of the scalar contribution $\mu_{SC.ap}$ is:

$$\mu_{SC.ap}(\dot{\gamma}) = \begin{cases} \mu_{SC.M.ap} = \eta_P + \frac{\tau_{printed}}{\dot{\gamma}_c}, & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \eta_P + \frac{\tau_{printed}}{\dot{\gamma}}, & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (2)$$

where $\mu_{SC.M.ap}$ is the maximum apparent viscosity contribution from the scalar. Thus, the deposited material has two viscosity contributions, one from the scalar (Eq. 2) and the other from the material itself (Eq. 1). They are governed with a mixture viscosity approach:

$$\mu_{mix} = \frac{\rho\mu_{ap} + S_c\mu_{SC.ap}}{\rho + S_c} \quad (3)$$

Equation 3 represents that the already printed material carries a slightly larger viscosity $\mu_{mix} > \mu_{ap}$ for a constant scalar concentration $S_c > 0$ and the material being deposited has the viscosity $\mu_{mix} = \mu_{ap}$. To produce a large viscosity difference between the two layers, the following characterization is used:

$$\mu_{mix.ap}(\dot{\gamma}) = \begin{cases} \mu_{SC.ap}, & \text{for } \mu_{mix} > \mu_{ap} \\ \mu_{ap}, & \text{for } \mu_{mix} \leq \mu_{ap} \end{cases} \quad (4)$$

which means the viscosity of the bottom and top layer is presented by Eqs. 2 and 1, respectively. Equation 4 is used to calculate the deviatoric stress tensor S_T by using the generalized Newtonian fluid approach:

$$S_T = 2\mu_{mix.ap}(\dot{\gamma})D_T \quad (5)$$

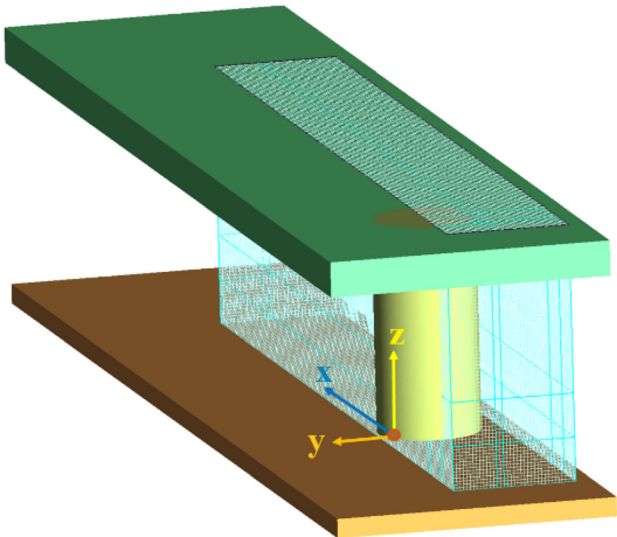


Fig. 1. Model geometry with the symmetrical computational domain.

where $\mathbf{D}_T = (1/2)((\nabla \mathbf{u}) + (\nabla \mathbf{u})')$ represents the deformation rate tensor and $\dot{\gamma} = \sqrt{2tr(\mathbf{D}_T^2)}$ is the strain rate.

Within the above assumptions, the flow dynamics is governed by the continuity and momentum equations:

$$\nabla \cdot \mathbf{u} = 0 \quad (6)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \rho \mathbf{g} + \nabla \cdot \mathbf{S}_T \quad (7)$$

where $\mathbf{g} = (0, 0, -g)$ is the gravitational acceleration vector, $\mathbf{u}$ is the velocity vector, t is the time, and p is the pressure. The properties of the material used in the CFD model are presented in Table I.

Equations 6 and 7 are solved numerically with the finite volume technique, using the commercial CFD software FLOW-3D[®]. The viscous stress, pressure and velocity components are solved implicitly. The momentum advection is calculated explicitly with second-order accuracy. Furthermore, the free surface tracking is done by the Volume-of-Fluid technique; see details in Refs. 24 and 25. Figure 2 visualizes the dynamic viscosity of the CFD model. The longitudinal section (top) and cross section (bottom) are taken through the nozzle diameter with side and front views, respectively. The model is simulated with $\tau_{printing} = 100 \text{ Pa}$ and $\tau_{printed} = 400 \text{ Pa}$. It can be seen that the model is able to predict different viscosity distributions in different layers, where red and greenish-blue colors respectively represent the maximum apparent viscosity of the bottom and top layer.

RESULTS AND DISCUSSION

In Figs. 3, 4, and 5, the developed CFD model is executed with different values of yield stress of the printed layer $\tau_{printed}$, which is presented in terms of the viscosity distributions, cross sections, deformation of bottom layer and extrusion pressure. The cross-sectional shapes are plotted at the middle of the length of the printed layer. The bottom layer deformation represents the percentile relative width of the bottom layer after printing the next layer. The extrusion pressure represents the pressure at the nozzle exit that is required to spread the

printing material with shear deformation. Note that the hydrostatic pressure for a single layer is approximately 123 Pa.

The viscosity distributions for different values of $\tau_{printed}$ are presented in Fig. 3. It shows that the viscosity profiles of the printing layer are similar to each other, whereas those of the printed layer are different for $\tau_{printed} = 100, 200$, and 400 Pa . In the case of $\tau_{printed} = \tau_{printing}$ (Fig. 3a), most of the material of the printed layer underneath the nozzle is found yielded (blue) because an effective decrease in viscosity is observed because of the total pressure, which includes both the hydrostatic and extrusion pressure. The yielded zone reduces when increasing the $\tau_{printed}$ (Fig. 3b and c). This is because the increased yield stress permits a higher load before yielding.

The impact of $\tau_{printed}$ on the cross-sectional shapes is presented in Fig. 4. It shows that the bottom layer deforms less with increasing $\tau_{printed}$. Furthermore, it

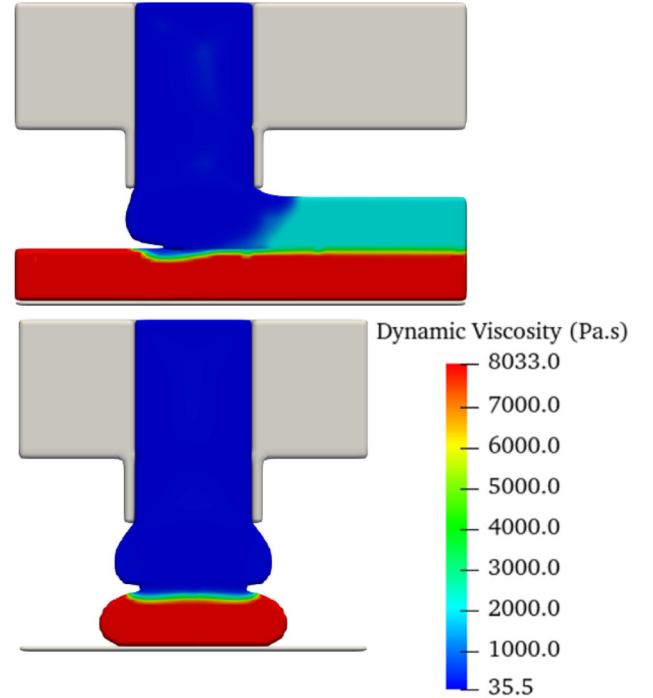


Fig. 2. Viscosity evaluation of the CFD model. Moreover, the yield stress of the bottom layer is 400 Pa while the top layer is printing with a yield stress of 100 Pa.

Table I. Material properties

Parameters	Units	Values
Density, ρ	kg.m^{-3}	1000
Dynamic yield stress of printing layer, $\tau_{printing}$	Pa	100
Dynamic yield stress of printed layer, $\tau_{printed}$	Pa	100–1500
Plastic viscosity, η_p	Pa.s	33
Critical shear rate, $\dot{\gamma}_c$	s^{-1}	0.05

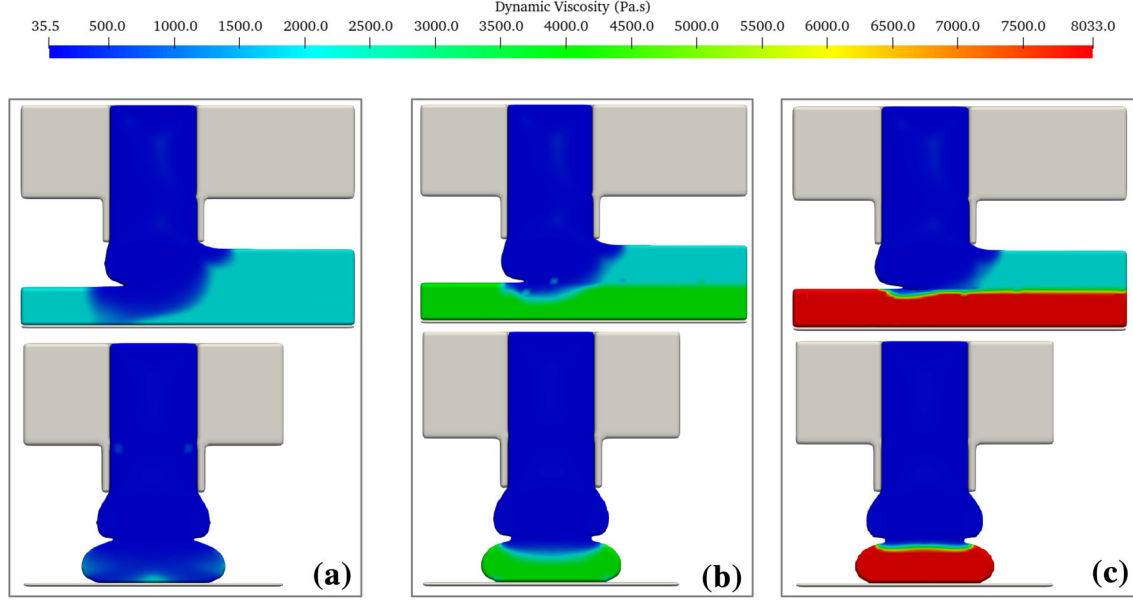


Fig. 3. Effect of the yield stress of printed material $\tau_{printed}$ on the viscosity distributions, where $\tau_{printing} = 100$ Pa. (a) $\tau_{printed} = 100$ Pa; (b) $\tau_{printed} = 200$ Pa; (c) $\tau_{printed} = 400$ Pa.

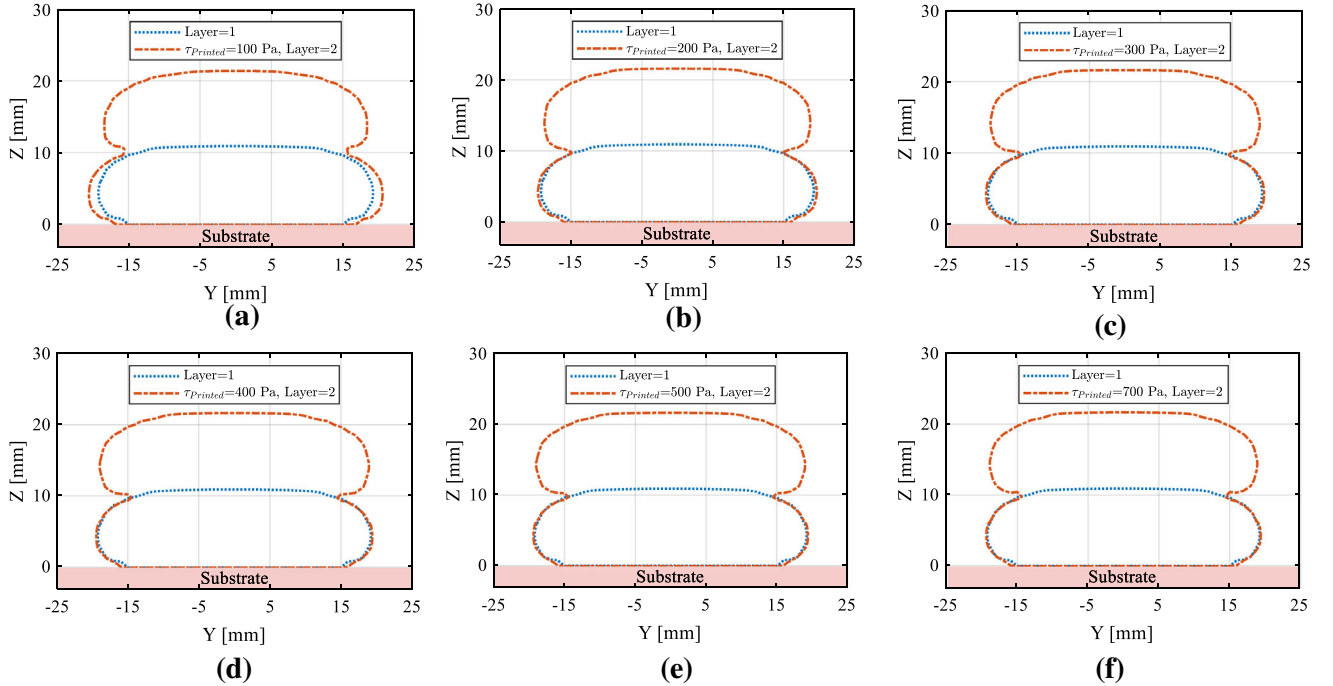


Fig. 4. Effect of the yield stress of printed material $\tau_{printed}$ on the cross-sectional shapes, where $\tau_{printing} = 100$ Pa. (a) $\tau_{printed} = 100$ Pa; (b) $\tau_{printed} = 200$ Pa; (c) $\tau_{printed} = 300$ Pa; (d) $\tau_{printed} = 400$ Pa; (e) $\tau_{printed} = 500$ Pa; (f) $\tau_{printed} = 700$ Pa.

is found that most of the deformation can be eliminated by increasing $\tau_{printed}$ to the double of $\tau_{printing}$ (Fig. 4b). However, further increase in $\tau_{printed}$ results in a little improvement in deformation, and for $\tau_{printed} \geq 400$ Pa, the cross-sectional shapes of bottom layer seem to coincide (Fig. 4d, e and f). The amount of deformation that occurred at the bottom layer is quantified in Fig. 5a, and for $\tau_{printed} \geq 400$ Pa, the deformation is $< 1\%$.

The extrusion pressure is presented in Fig. 5b as a function of the yield stress of the printed layer. The extrusion pressure during printing of the bottom layer is represented by the straight line that has no influence for different $\tau_{printed}$. The maximum extrusion pressure is observed for the bottom layer that is printed into the non-deformable solid substrate, whereas for the other layer, the extrusion pressure varies depending on the

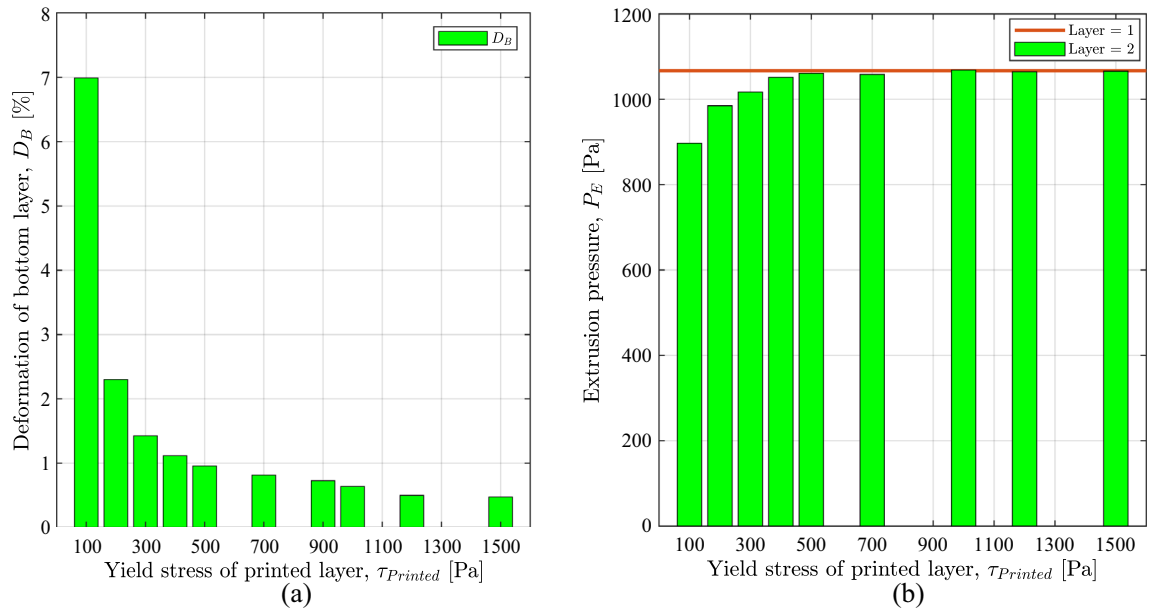


Fig. 5. (a) Deformation of bottom layer and (b) extrusion pressure as a function of the yield stress of printed material $\tau_{printed}$, where $\tau_{printing} = 100$ Pa.

deformability of the printed layer. The more the printed layer deforms, the more the effective distance between the nozzle exit and the printed layer increases; hence, less extrusion pressure is observed. Moreover, for $\tau_{printed} \geq 400$ Pa the extrusion pressure starts to converge, which is because the effective distance does not change.

CONCLUSION

A CFD model was developed to predict the extrusion-deposition flow of the viscoplastic material that develops its properties over time. The materials property requires two different yield stress approaches (for the printed and printing layers) that were modeled with the Bingham constitutive equations. The CFD model was found capable of differing the viscosity between the fluid layers precisely as formulated. We numerically analyzed the influence of yield stress buildup on the bottom layer deformation. Within the investigated parameters, the preliminary outcomes showed that most of the deformation could be eliminated by having the bottom layer develop the yield stress to twice the value of the printing material. Thus, this study showcases that CFD models have a potential to assist in finding the optimal material development that leads to stable MEX-AM printing.

ACKNOWLEDGEMENTS

The authors acknowledge the support from the Danish Council for Independent Research (DFF) | Technology and Production Sciences (FTP) (Contract No. 8022-00042B). Moreover, the authors

thank FLOW-3D® for their support regarding licenses.

CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

REFERENCES

1. Z. Chen, Z. Li, J. Li, C. Liu, C. Lao, Y. Fu, C. Liu, Y. Li, P. Wang, and Y. He, *J. Eur. Ceram. Soc.* 39, 661 (2019).
2. J. Lindahl, A. A. Hassen, S. Romberg, B. Hedger, P. Hedger Jr, M. Walch, T. DeLuca, W. Morrison, P. Kim, A. Roschli, D. Nuttal, J. Czachowski, B. Post, L. Love, and V. Kunc, in *United States (Oak Ridge National Lab. (ORNL), Oak Ridge, TN (US))*, (2018).
3. V. Mechtcherine, F.P. Bos, A. Perrot, W.R.L. da Silva, V.N. Nerella, S. Fataei, R.J.M. Wolfs, M. Sonebi, and N. Roussel, *Cem. Concr. Res.* 132, 106037 (2020).
4. J. Jiang, and Y. Ma, *Micromachines* 11, 633 (2020).
5. T.T. Le, S.A. Austin, S. Lim, R.A. Buswell, R. Law, A.G.F. Gibb, and T. Thorpe, *Cem. Concr. Res.* 42, 558 (2012).
6. O.A. Mohamed, S.H. Masood, and J.L. Bhowmik, *Addit. Manuf.* 18, 164 (2017).
7. H. Xia, J. Lu, and G. Tryggvason, *Comput. Methods Appl. Mech. Eng.* 346, 242 (2019).
8. H. Xia, J. Lu, and G. Tryggvason, *Int. J. Heat Mass Transf.* 138, 1310 (2019).
9. M. Jabbari, I. Baran, S. Mohanty, R. Comminal, M.R. Sonne, M.W. Nielsen, J. Spangenberg, and J.H. Hattel, *Adv. Mech. Eng.* 10, 5 (2018).
10. R. Comminal, M.P. Serdeczny, D.B. Pedersen, and J. Spangenberg, *Addit. Manuf.* 29, 100753 (2019).
11. R. Comminal, M. P. Serdeczny, D. B. Pedersen, and J. Spangenberg, in *Proc. Annu. Int. Solid Free. Fabr. Symp.* 1855 (2018).
12. M. P. Serdeczny, R. Comminal, D. B. Pedersen, and J. Spangenberg, in *Proc. Annu. Int. Solid Free. Fabr. Symp.* 1849 (2018).
13. M.P. Serdeczny, R. Comminal, D.B. Pedersen, and J. Spangenberg, *Addit. Manuf.* 28, 419 (2019).

14. R. Comminal, M.P. Serdeczny, D.B. Pedersen, and J. Spangenberg, *Addit. Manuf.* 20, 68 (2018).
15. R. Comminal, M. P. Serdeczny, D. B. Pedersen, and J. Spangenberg, in *18th Int. Conf. Eur. Soc. Precis. Eng. Nanotechnol. Euspen 18*, (2018).
16. M. P. Serdeczny, R. Comminal, D. B. Pedersen, and J. Spangenberg, in *18th Int. Conf. Eur. Soc. Precis. Eng. Nanotechnol. Euspen 18*, (2018).
17. R. Comminal, W.R. Leal da Silva, T.J. Andersen, H. Stang, and J. Spangenberg, *Cem. Concr. Res.* 138, 106256 (2020).
18. N. Roussel, J. Spangenberg, J. Wallevik, and R. Wolfs, *Cem. Concr. Res.* 135, 106075 (2020).
19. J. Spangenberg, W.R. Leal da Silva, R. Comminal, M.T. Mollah, T.J. Andersen, and H. Stang, *RILEM Tech. Lett.* 6, 119 (2021).
20. M.T. Mollah, R. Comminal, M.P. Serdeczny, D.B. Pedersen, and J. Spangenberg, *Addit. Manuf.* 46, 102193 (2021).
21. E.J. O'Donovan, R.I. Tanner, and J. Non-Newton, *Fluid Mech.* 15, 75 (1984).
22. E. C. Bingham, *An Investigation of the Laws of Plastic Flow* (US Government Printing Office, 1917).
23. FLOW-3D, "Solving The World's Toughest CFD Problems", (FLOW Sci. (n.d.)). <https://www.flow3d.com/products/flow-3d/> (accessed May 10, 2020).
24. C.W. Hirt, and B.D. Nichols, *J. Comput. Phys.* 39, 201 (1981).
25. R. Comminal, J. Spangenberg, and J.H. Hattel, *J. Comput. Phys.* 283, 582 (2015).

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.